EP Curs6 – Comparing Systems

Comparing computer systems using sample data

Overview

- Sample vs Population
- Confidence interval for the mean
- Approximate visual test
- One sided confidence intervals
- Sample size for determining mean and proportions

Sample vs Population

- Generate several million random numbers with mean μ and standard deviation σ
  Select a sample of n observations

$$\overline{x} \neq \mu$$

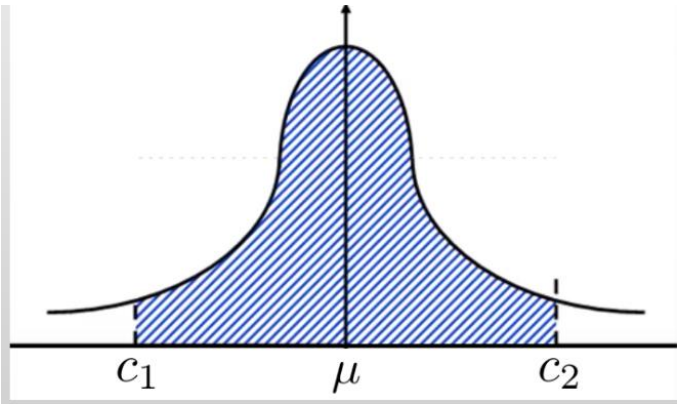
- Sample mean <> Population mean
- Population (unknown) - greek letters
  - Population characteristics —> parameters
- Sample - english letters
  - Sample estimates —> statistics

Confidence interval for the mean

- k samples => k sample means
  - Cannot get a single estimate of μ
  - Use bounds c1 and c2

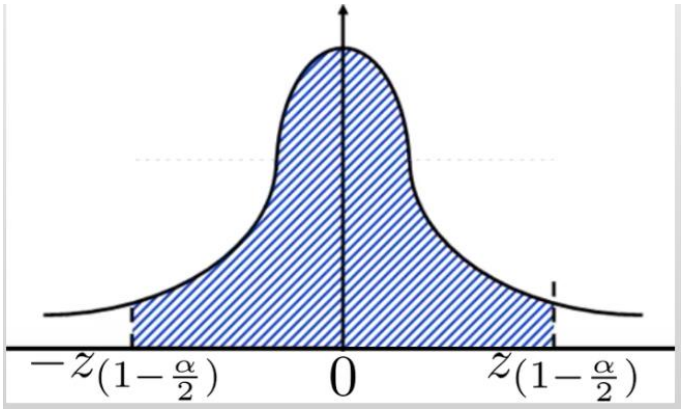
$$Probability\{c_1 \leq \mu \leq c_2\} = 1 - \alpha$$

- Confidence interval: (C1, C2)
- Significance level: α
- Confidence level: 100(1-α)
- Confidence coefficient: 1



Confidence interval

- 90% confidence interval
  - 5-percentile and 95-percentile of the sample means
- Central limit theorem
  - Sample mean is normally distributed for large sample sizes, regardless of the distribution from which we are sampling.



$$\overline{x} \sim N(\mu, \sigma/\sqrt{n})$$

Where μ = population mean and σ = population standard deviation

- Standard error: Standard deviation of the sample mean $s = \sigma/\sqrt{n}$
- 100(1-α)% confidence interval for μ:

$$(\overline{x} - z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}}, \overline{x} + z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}})$$

$$z_{1-\frac{\alpha}{2}} = (1-\frac{\alpha}{2}) - quantile\ \ of\ \ N(0,1)$$

Example

$$\overline{x} = 3.90,\ \ s = 0.95,\ \ n = 32$$

- 90% confidence interval for the mean

$$(\overline{x} - z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}}, \overline{x} + z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}})$$

$$3.90 \mp (1.645)(0.95)/\sqrt{32} = (3.62, 4.17)$$

- 95% confidence interval for the mean

$$3.90 \mp (1.960)(0.95)/\sqrt{32} = (3.57, 4.23)$$

- 99% confidence interval for the mean

$$3.90 \mp (2.576)(0.95)/\sqrt{32} = (3.46, 4.33)$$

TABLE A.2 Quantiles of the Unit Normal Distribution

p	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.5	0.000	0.025	0.050	0.075	0.100	0.126	0.151	0.176	0.202	0.228
0.6	0.253	0.279	0.305	0.332	0.358	0.385	0.412	0.440	0.468	0.496
0.7	0.524	0.553	0.583	0.613	0.643	0.674	0.706	0.739	0.772	0.806
0.8	0.842	0.878	0.915	0.954	0.994	1.036	1.080	1.126	1.175	1.227

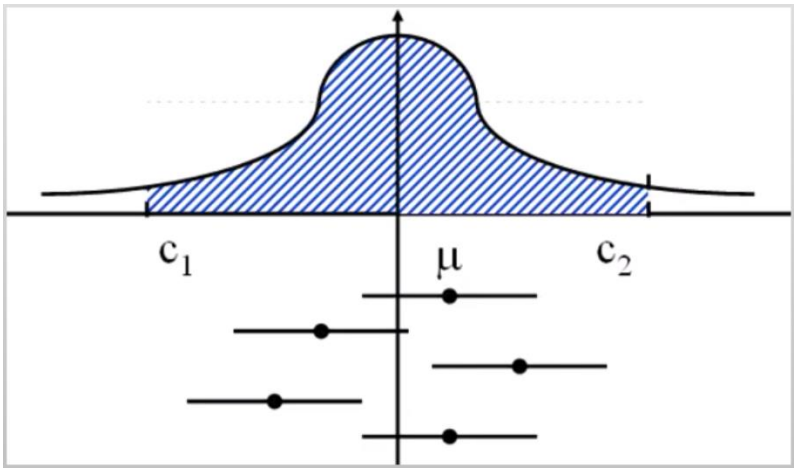
p	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.90	1.282	1.287	1.293	1.299	1.305	1.311	1.317	1.323	1.329	1.335
0.91	1.341	1.347	1.353	1.359	1.366	1.372	1.379	1.385	1.392	1.398
0.92	1.405	1.412	1.419	1.426	1.433	1.440	1.447	1.454	1.461	1.468
0.93	1.476	1.483	1.491	1.499	1.506	1.514	1.522	1.530	1.538	1.546
0.94	1.555	1.563	1.572	1.580	1.589	1.598	1.607	1.616	1.626	1.635
0.95	1.645	1.655	1.665	1.675	1.685	1.695	1.706	1.717	1.728	1.739
0.96	1.751	1.762	1.774	1.787	1.799	1.812	1.825	1.838	1.852	1.866
0.97	1.881	1.896	1.911	1.927	1.943	1.960	1.977	1.995	2.014	2.034
0.98	2.054	2.075	2.097	2.120	2.144	2.170	2.197	2.226	2.257	2.290

p	0.0000	0.0001	0.0002	0.0003	0.0004	0.0005	0.0006	0.0007	0.0008	0.0009
0.990	2.326	2.330	2.334	2.338	2.342	2.346	2.349	2.353	2.357	2.362
0.991	2.366	2.370	2.374	2.378	2.382	2.387	2.391	2.395	2.400	2.404
0.992	2.409	2.414	2.418	2.423	2.428	2.432	2.437	2.442	2.447	2.452
0.993	2.457	2.462	2.468	2.473	2.478	2.484	2.489	2.495	2.501	2.506
0.994	2.512	2.518	2.524	2.530	2.536	2.543	2.549	2.556	2.562	2.569
0.995	2.576	2.583	2.590	2.597	2.605	2.612	2.620	2.628	2.636	2.644
0.996	2.652	2.661	2.669	2.678	2.687	2.697	2.706	2.716	2.727	2.737
0.997	2.748	2.759	2.770	2.782	2.794	2.807	2.820	2.834	2.848	2.863
0.998	2.878	2.894	2.911	2.929	2.948	2.968	2.989	3.011	3.036	3.062
0.999	3.090	3.121	3.156	3.195	3.239	3.291	3.353	3.432	3.540	3.719

(asta e pt 90%, se ia procentul droit cu radical si se ia din table nr)

Confidence interval

- Take 100 samples and construct confidence interval for each sample, the interval would include the population mean in 90 cases.



Confidence interval for small samples

- How CI works - assume large number of observations => mean is normally distributed
- Large sample = 30
    - Not absolute bound
    - Exceptions exist (30+ observations and nowhere close to a normal distribution)
- What can you do if you do not have 30 observations?
- 100(1-α)% confidence interval for n<30

$$(\overline{x} - t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}}, \overline{x} + t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}})$$

- $t_{[1-\frac{\alpha}{2};n-1]} = (1-\frac{\alpha}{2})$ - quantile of a t-variable with n-1 degrees of freedom

Example

- Sample: -0.04, -0.19, 0.14, -0.09, -0.14, 0.19, 0.04, 0.09
- Mean = 0
- Sample standard deviation = 0.138
- For 90% interval: $t_{[1-\frac{\alpha}{2};n-1]} = t_{[0.95;7]} =$

$$(\overline{x} - t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}}, \overline{x} + t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}})$$

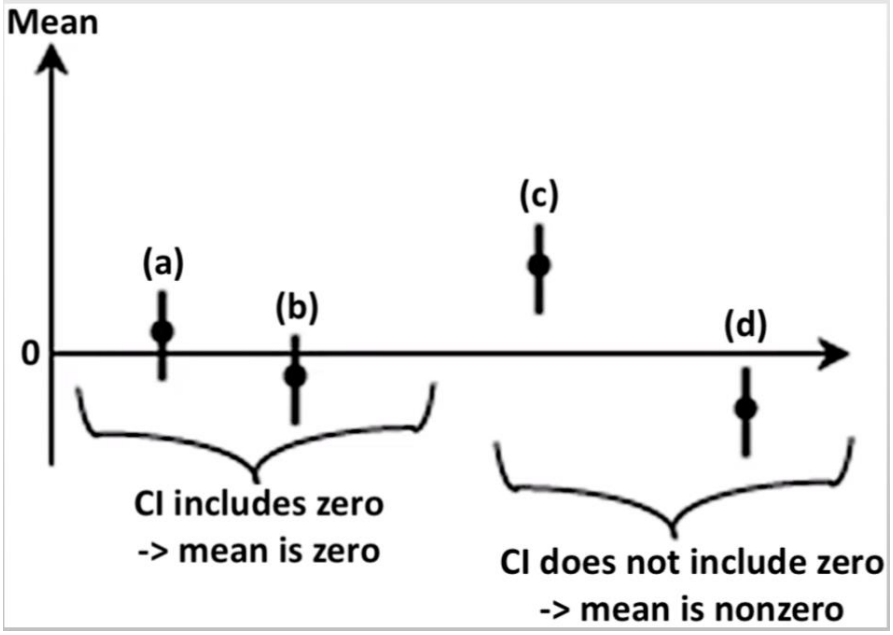
- Confidence interval for the mean:

$$0 \mp 1.895 \times 0.138/\sqrt{8} = 0 \mp 0.092 = (-0.092, 0.092)$$

TABLE A.4 Quantiles of the *t* Distribution

	p							
n	**0.6000**	**0.7000**	**0.8000**	**0.9000**	**0.9500**	**0.9750**	**0.9950**	**0.9995**
1	0.325	0.727	1.377	3.078	6.314	12.706	63.657	636.619
2	0.289	0.617	1.061	1.886	2.920	4.303	9.925	31.599
3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.924
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	4.032	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.807	3.768
24	0.256	0.531	0.857	1.318	1.711	2.064	2.797	3.745
25	0.256	0.531	0.856	1.316	1.708	2.060	2.787	3.725
26	0.256	0.531	0.856	1.315	1.706	2.056	2.779	3.707
27	0.256	0.531	0.855	1.314	1.703	2.052	2.771	3.690
28	0.256	0.530	0.855	1.313	1.701	2.048	2.763	3.674
29	0.256	0.530	0.854	1.311	1.699	2.045	2.756	3.659
30	0.256	0.530	0.854	1.310	1.697	2.042	2.750	3.646
60	0.254	0.527	0.848	1.296	1.671	2.000	2.660	3.460
90	0.254	0.526	0.846	1.291	1.662	1.987	2.632	3.402
120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.373

Testing for zero mean



Example

- Difference in processor times: {1.5, 2.6, -1.8,1.3, -0.5, 1.7, 2.4}
- Can we say with 99% confidence that one is superior to the other?
  Mean = 7.2 / 7 = 1.03
  Sample size = n = 7
  Sample variance = (22.84 - 7.2 * 7.2 / 7) / 6 = 2.57
  Sample standard deviation = sqr(2.57) = 1.6
- Confidence interval:

$$1.03 \mp t * 1.6/\sqrt{7} = 1.03 \mp 0.6t$$
$$100(1-\alpha) = 99,\ \ \alpha = 0.01,\ \ 1-\alpha/2 = 0.995,\ \ t_{[0.995,6]} = 3.707$$

TABLE A.4 Quantiles of the *t* Distribution

	p							
n	**0.6000**	**0.7000**	**0.8000**	**0.9000**	**0.9500**	**0.9750**	**0.9950**	**0.9995**
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3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.924
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
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6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
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27	0.256	0.531	0.855	1.314	1.703	2.052	2.771	3.690
28	0.256	0.530	0.855	1.313	1.701	2.048	2.763	3.674
29	0.256	0.530	0.854	1.311	1.699	2.045	2.756	3.659
30	0.256	0.530	0.854	1.310	1.697	2.042	2.750	3.646
60	0.254	0.527	0.848	1.296	1.671	2.000	2.660	3.460
90	0.254	0.526	0.846	1.291	1.662	1.987	2.632	3.402
120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.373

- 99% confidence interval = (-1.21, 3.27)
- Opposite signs => we cannot say with 99% confidence that the mean difference is significantly different from zero.
- Answer: They are the same.
- Answer: The difference is zero.
- Difference in processor times: {1.5, 2.6, -1.8, 1.3, -0.5, 1.7, 2.4}

- 99% confidence interval = (-1.21, 3.27)
- Is the difference 1?
- Answer: Yes, the difference is 1.

Paired vs unpaired comparisons

- Paired
  - One-to-one correspondence between the ith test on system A and ith test on system B
  - Example: performance on i-th workload
  - Use confidence interval of the difference
- Unpaired
  - No correspondence
  - Number of observations may differ
  - Example
    - n people on system A, n on system B
    - Need more sophisticated method

Example

- Performance: {(5.4, 19.1), (16.6, 3.5), (0.6, 3.4), (1.4, 2.5), (0.6, 3.6) (7.3, 1.7)}
  Is one system better?
- Differences: {-13.7, 13.1, -2.8, -1.1, -3.0, 5.6}
  Sample mean = -0.32
  Sample variance = 81.62
  Sample standard deviation = 9.03
  Confidence interval for the mean:

$$-0.32 \mp t\sqrt{81.62/6} = -0.32 \mp t(3.69), \;\; t_{[0.95,5]} = 2.015$$

$$\text{90\% confidence interval:} \quad -0.32 \mp (2.015)(3.69) = (-7.75, 7.11)$$

- Answer: No, they are not different.

Unpaired observations:

- Compute sample means

$$\overline{x}_a = \frac{1}{n_a}\sum_{i=1}^{n_a} x_{ia}$$

$$\overline{x}_b = \frac{1}{n_b}\sum_{i=1}^{n_b} x_{ib}$$

- Compute the sample standard deviations

$$s_a = \left\{ \frac{(\sum_{i=1}^{n_a} x_{ia}^2) - n_a x_a^{-2}}{n_a - 1} \right\}^{\frac{1}{2}}$$

$$s_b = \left\{ \frac{(\sum_{i=1}^{n_b} x_{ib}^2) - n_b x_b^{-2}}{n_b - 1} \right\}^{\frac{1}{2}}$$

- Compute mean differences

$$(\overline{x}_a - \overline{x}_b)$$

- Compute standard deviation of the mean differences

$$s = \sqrt{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b} \right)}$$

- Compute the effective number of degrees of freedom

$$\nu = \frac{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b} \right)^2}{\frac{1}{n_a+1} \left(\frac{s_a^2}{n_a} \right)^2 + \frac{1}{n_b+1} \left(\frac{s_b^2}{n_b} \right)^2} - 2$$

- Compute the confidence interval for the mean differences

$$(\overline{x}_a - \overline{x}_b) \mp t_{[1-\alpha/2;\nu]} s$$

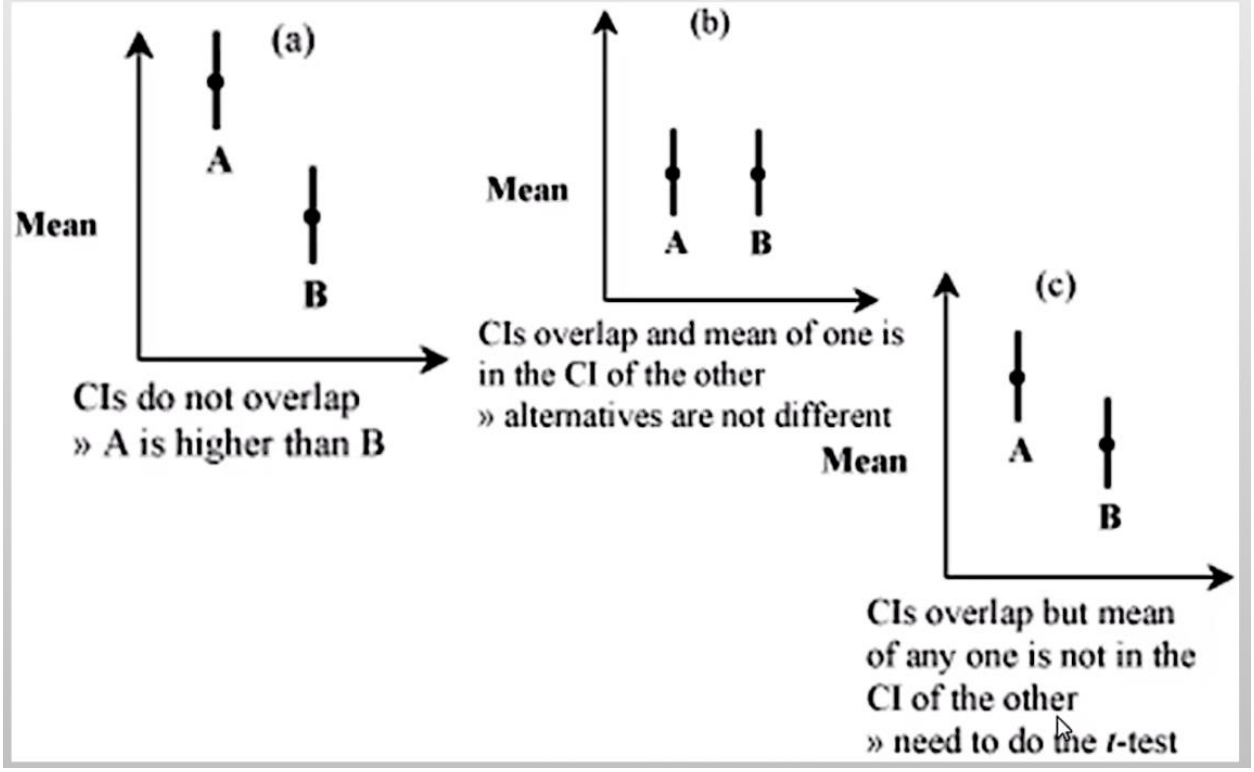
Example

- Times on system A: {5.36, 16.57, 0.62, 1.41, 0.64, 7.26}
  Times on system B: {19.12, 3.52, 3.38, 2.50, 3.60, 1.74}
- Are the two systems significantly different?

- System A
  - Mean $\overline{x}_a = 5.31$
  - Variance $s_a^2 = 37.92$
  - $n_a = 6$
- System B
  - Mean $\overline{x}_b = 5.64$
  - Variance $s_b^2 = 44.11$
  - $n_b = 6$

- Mean difference xa — xb = -0.33
- Standard deviation of the mean difference = 3.698
- Effective number of degrees of freedom *f* = 11.921
- 0.95-quantile of a t-variate with 12 degrees of freedom = 1.71
- 90% confidence interval for the difference = (-6.92, 6.26)
- The confidence interval includes zero => the two systems are not different

Approximate visual test



Example

- Times on system A: {5.36, 16.57, 0.62, 1.41, 0.64, 7.26}
  Times on system B: {19.12, 3.52, 3.38, 2.50, 3.60, 1.74}
  $$t_{[0.95,5]} = 2.015$$
- 90% confidence interval for the mean of A
  $$5.31 \mp (2.015)\sqrt{37.92/6} = (0.24, 10.38)$$
- 90% confidence interval for the mean of B
  $$5.64 \mp (2.015)\sqrt{44.11/6} = (0.18, 11.10)$$

- Confidence intervals overlap and the mean of one falls in the confidence interval of the other =>
- The two systems are not different at this level of confidence

What level of confidence to use?

- 90%, 95%, 99% ?
- Base on the loss that you will sustain if the parameter is outside the range and the gain you would have if the parameter is inside the range.
- Low loss -> low confidence is fine
    - Example: Lottery of $5million with probability 10^(-7)
        - 90% confidence —> buy 9 million tickets
        - 0.01% confidence level if fine
    - 50% confidence may or may not be too low
    - 99% confidence may or may not be too high

One-sided confidence intervals

- Two side intervals - 90% confidence
    - P(difference > upper limit) = 5%
    - P(difference < lower limit) = 5%
- One-sided confidence interval
    - One sided question: Is the mean greater than 0?
- One-sided lower confidence interval for μ

$$\left(\overline{x} - t_{[1-\alpha;n-1]}\frac{s}{\sqrt{n}}, \overline{x}\right)$$

Note t at 1-α (not at 1-α/2)

- One-sided upper confidence interval for μ

$$\left(\overline{x}, \overline{x} + t_{[1-\alpha;n-1]}\frac{s}{\sqrt{n}}\right)$$

- For large samples - use z instead of t

Example

- Time between crashes

System	Number	Mean	Stdv
A	972	124.10	198.20
B	153	141.47	226.11

- Assume unpaired observations
- Mean difference

$$\overline{x}_A - \overline{x}_B = 124.10 - 141.47 = -17.37$$

- Standard deviation of the difference

$$s = \sqrt{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}\right)} = \sqrt{\left(\frac{(198.20)^2}{972} + \frac{(226.11)^2}{153}\right)} = 19.35$$

- Effective number of degrees of freedom

$$\nu = \frac{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}\right)^2}{\frac{1}{n_a+1}\left(\frac{s_a^2}{n_a}\right)^2 + \frac{1}{n_b+1}\left(\frac{s_b^2}{n_b}\right)^2} - 2 = \ldots = 191.05$$

- v > 30 use z instead of t
- One-sided test => Use $z_{0.90}$ = 1.28 for 90% confidence
- 90% confidence interval: (-17.37, -17.37+1.28*19.35) = (-17.37, 7.402)
- CI includes zero => system A is not more susceptible to crashes than system B.

Sample size for determining mean

- Large sample => narrower confidence interval. - Higher confidence
- Question
  - How many observations n to get an accuracy of +/- r% and a confidence level of 100(1-α)%?

$$\overline{x} \mp z\frac{s}{\sqrt{n}}$$

- r% accuracy

$$CI = (\overline{x}(1 - r/100), \overline{x}(1 + r/100))$$

$$\overline{x} \mp z\frac{s}{\sqrt{n}} = \overline{x}\left(1 \mp \frac{r}{100}\right) \Rightarrow z\frac{s}{\sqrt{n}} = \overline{x}\frac{r}{100} \Rightarrow n = \left(\frac{100zs}{r\overline{x}}^2\right)$$

Example

- Sample mean of the response time = 20 sec
- Sample standard deviation = 5
- Question: How many repetitions are needed to get the response time accurate within 1 second at 95% confidence?
- Required accuracy is 1 in 20 = 5%

$$\overline{x} = 20, \quad s = 5, \quad z = 1.960, \quad r = 5$$

$$n = \left(\frac{100 \times 1.960 \times 5}{5 \times 20}\right)^2 = 9.8^2 = 96.04$$

- Answer: 97 observations are needed

Takeaways

- All statistics based on a sample are random and should be specified with a confidence interval
- If the confidence interval includes zero, the hypothesis that the population mean is zero cannot be rejected
- Paired observations => test the difference for zero mean
- Unpaired observations => more sophisticated